# 1. Identity

| | |
|---|---|
| Document | STAR-PKG-STAR-PRECESSION-EN · v2.0 · 2026-10-21 |
| Package | star-precession |
| Version | 0.1.0 |
| Capability | star.frames.precession-iau76 |
| Maturity | RELEASED |
| Python | ⩾3.9 |
| Source commit | 94fcf7c |
| Source | https://github.com/Andrea07072000/S.T.A.R.-/tree/main/packages/star_precession |

# 2. Licence and obligations

Apache License 2.0. You may use, copy, modify and distribute it, also commercially, and you receive a patent licence from the contributors. When you redistribute it: keep the LICENSE file and the copyright notices (and a NOTICE file, if a future version adds one), and state in modified files that you changed them. You may not use the S.T.A.R. name to endorse your product.

# 3. What it needs at run time

Nothing beyond the Python standard library: it works offline.

# 4. Integrity

SHA-256 fingerprints of the files published on this website. Check a downloaded file with:

```
python -c "import hashlib,sys;print(hashlib.sha256(open(sys.argv[1],'rb').read()).hexdigest())" star_precession-0.1.0-py3-none-any.whl
```

| FILE | SHA-256 |
|---|---|
| star_precession-0.1.0-py3-none-any.whl | 461606b45b709e7d491e50960d0cca319d0f882da7f2a0052c05535da884953c |
| star_precession.py | 5ed6d2722b7804a546434851509529d0197698a6a5cd04da6fdb4d94b650c938 |
| star_precession-0.1.0-source.zip | 07e43a5a52c2388e5bea7e8381dca91f4fd4205b528380f262abee8bbf152463 |

# 5. What it does, and its limits

IAU 1976 precession matrix and vector rotations between J2000 and the mean equator and equinox of date over 1800-2200; equal to ERFA pmat76 to 1.2e-16 and within 0.21 arcsec of PyEphem's own precession.

IAU 1976 only: not the IAU 2006 precession (they differ by about 0.3 arcsec per century), no nutation (up to 17 arcsec), no frame bias between J2000 and GCRF (17 milliarcseconds). Enough for vectors known to a hundredth of a degree; not for arcsecond astrometry.

- `precession_matrix(jd_day, jd_frac=0)`: P = R3(-z) R2(theta) R3(-zeta) with the Lieske (1979) angles; identity at J2000.0. Reproduces the SOFA validation values of `pmat76` to 1e-15.
- `mod_from_j2000(r, ...)`, `j2000_from_mod(r, ...)` (the transpose) and `precession_angles_arcsec`.
- Measured on 500 dates over 1800-2200: every matrix element within 1.2e-16 of ERFA; 1500 directions within 0.21 arcsec of PyEphem, which carries its own precession code.
- Every function returns finite floats or raises `ValueError`: non-numeric, boolean, NaN or infinite input, a vector that is not three finite numbers, a date outside 1800-01-01 to 2200-01-01.

# 6. Export, use and certification

The package is published and freely available. That usually takes software out of export-control lists, but not every use of it: read the page on space activities and export control before using it in a controlled programme. It is not certified or qualified for flight; the evidence describes what was tested and how.

# 7. Citation and bill of materials

```
@software{cavazzini_star_precession_0_1_0,
  author  = {Cavazzini, Andrea},
  title   = {{star-precession}: Between J2000 and the mean equator of date (IAU 1976)},
  version = {0.1.0},
  year    = {2026},
  url     = {https://andrea07072000.github.io/S.T.A.R.-/software/star_precession/},
  note    = {Apache-2.0. Source: https://github.com/Andrea07072000/S.T.A.R.-, commit 94fcf7c}
}
```

CITATION.cff · BibTeX · SBOM CycloneDX 1.5 · SBOM SPDX 2.3

# 8. Warranty and liability

Provided "as is", without warranties or conditions of any kind (Apache License 2.0, sections 7 and 8), to the extent the law allows. This never excludes liability that the law does not allow to exclude.

Related: space activities and export control · acceptable use · evidence and limits · licence texts